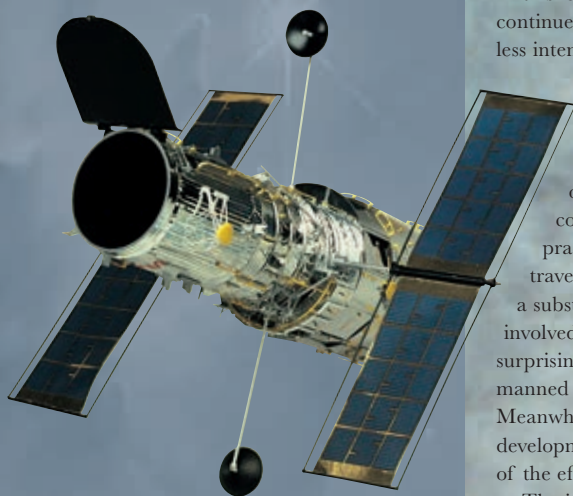




“Aiming at the stars ... is a problem to occupy generations, so that no matter how much progress one makes, there is always the thrill of just beginning.”

ROBERT GODDARD

LETTER TO H.G. WELLS, 1932



HUBBLE TELESCOPE

The Hubble Space Telescope (HST) was launched in April 1990. It is capable of observing objects far more clearly than most of Earth's largest optical telescopes. The HST was launched and has been repaired by Shuttle astronauts.

SHUTTLE TO SPACE

AFTER THE ENDING OF THE APOLLO PROGRAM, THE FOCUS OF MANNED SPACE FLIGHT SHIFTED TO REUSABLE SPACECRAFT AND SPACE STATIONS

IN THE 1970s manned space flight programs had to adapt to a rapidly changing situation that threatened their survival. The political heat had gone out of the competition between the superpowers which had energized the space race. Caught up in post-Vietnam and Watergate traumas, the United States was in one of its least bullish and assertive national moods. The world economy was also in crisis after decades of rapid expansion. The new demand was for a focus on cost-effectiveness and practical applications of space technology. In this context, the growing everyday usefulness of satellites and the spectacular triumphs of unmanned space probes inevitably led to a questioning of the need for manned flights at all.

Practical progress

In this less supportive context, space agencies continued with manned programs that were far less intensive in the short term, but still based on an ambitious long-term vision of a developing human presence in space. The logic of the development of a reusable space vehicle seemed overwhelming in the light of the expected cost advantage it should bring, and its practicality – it was hard to imagine space travel ever growing into a phenomenon on a substantial scale if every return to Earth involved being picked up from the sea. Not surprisingly, then, the focus of the American manned space program became the Shuttle. Meanwhile the Soviets took the lead in the development of space stations and exploration of the effects of long-duration space flights.

The idea of permanently inhabited space stations in orbit around the Earth had occurred to some of the earliest space-flight visionaries. In 1923 German rocket enthusiast Hermann Oberth wrote of an orbiting “observing station” from which astronauts with “powerful instruments”

SHUTTLE PILOTS

John Young, a veteran of the Apollo programme, and Robert Crippen were the astronauts at the controls of the Shuttle Columbia on its first flight into space on April 12, 1981.



would be able to see “fine detail on Earth” and give warning of icebergs on the oceans – thus averting any repetition of the Titanic disaster.

Oberth imagined this information being flashed to Earth in code by reflecting sunlight off a hand mirror.

As a disciple of Oberth, Wernher von Braun picked up an early enthusiasm for space stations that endured throughout his life. He drew up a blueprint for one in the 1950s and promoted the idea as a future follow-up to the Moon program during the second half of the 1960s. But it was the Soviets who first put a functioning space station in orbit, and who were to make space stations their most distinctive contribution to late 20th-century space exploration.

By 1969, with the goal of being first to the Moon beyond their grasp, and no desire